

Analysis of Algorithms (BDSF25M/A)

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Quiz 02

Time: 15 min

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Grading	CLO-1	CLO-2	CLO-3	CLO-4
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Read the guidance (similar as in class session) provided on page overleaf and solve following recurrences with only necessary steps. Note: in (1) 1 is n^0 and in (ii) n is n^1

(1) $T(n) = 3T\left(\frac{n}{3}\right) + 1$ $a=3, b=3, k=0$ (since $1 = n^0$)

Ratio: $\frac{a}{b^k} = \frac{3}{3^0} = \frac{3}{1} = 3$

Substitution: $T(n) = c \cdot \log_3 n + n^0 \sum_{j=0}^{\log_3 n - 1} (3)^j$

$T(n) = c \cdot n^1 + 1 \left(\frac{3^{\log_3 n} - 1}{3 - 1} \right)$ using geometric sum.

using:

$T(n) = c \cdot n^{\log_3 3} + n^k \sum_{j=0}^{\log_3 n - 1} \left(\frac{a}{b^k} \right)^j$

$T(n) = cn + \frac{n-1}{2}$

$T(n) = \Theta(n)$

(2) $T(n) = 5T\left(\frac{n}{2}\right) + n$

$a=5, b=2, k=1$ (n^1)

$\frac{a}{b^k} = \frac{5}{2^1} = 2.5$

$T(n) = c \cdot n^{\log_2 5} + n^1 \sum_{j=0}^{\log_2 n - 1} (2.5)^j$
 $= c \cdot n^{2.32} + n \left(\frac{2.5^{\log_2 n} - 1}{2.5 - 1} \right)$

$T(n) = cn^{2.32} + n \left(\frac{n^{\log_2 2.5} - 1}{1.5} \right)$

$T(n) = \Theta(n^{\log_2 5})$

(3) $T(n) = 3T\left(\frac{n}{2}\right) + n^2$

$a=3, b=2, k=2, \frac{a}{b^k} = \frac{3}{2^2} = \frac{3}{4} = 0.75$

$T(n) = c \cdot n^{\log_2 3} + n^2 \sum_{j=0}^{\log_2 n - 1} (0.75)^j$

Since ratio $0.75 < 1$ the summation converges to a constant $\sum_{j=0}^{\infty} (0.75)^j = \frac{1}{1-0.75} = 4$

$T(n) \propto c \cdot n^{1.58} + 4n^2$ So, $T(n) = \Theta(n^2)$

(4) $T(n) = 27T\left(\frac{n}{3}\right) + n^3$

$a=27, b=3, k=3$

$\frac{a}{b^k} = \frac{27}{3^3} = \frac{27}{27} = 1$

$T(n) = c \cdot n^{\log_3 27} + n^3 \sum_{j=0}^{\log_3 n - 1} (1)^j$

$= c \cdot n^3 + n^3 (\log_3 n)$

$T(n) = \Theta(n^3 \log n)$

Solving $T(n) = aT(n/b) + n^k$ via Backward Substitution

1. The Iterative Expansion

We start with the given recurrence and repeatedly substitute the definition of T into itself to observe the developing pattern.

Step 1:

$$T(n) = aT\left(\frac{n}{b}\right) + n^k$$

Step 2: Substitute $T\left(\frac{n}{b}\right) = aT\left(\frac{n}{b^2}\right) + \left(\frac{n}{b}\right)^k$:

$$T(n) = a\left[aT\left(\frac{n}{b^2}\right) + \left(\frac{n}{b}\right)^k\right] + n^k$$

$$T(n) = a^2T\left(\frac{n}{b^2}\right) + a\frac{n^k}{b^k} + n^k$$

Step 3: Substitute $T\left(\frac{n}{b^2}\right) = aT\left(\frac{n}{b^3}\right) + \left(\frac{n}{b^2}\right)^k$:

$$T(n) = a^2\left[aT\left(\frac{n}{b^3}\right) + \left(\frac{n}{b^2}\right)^k\right] + a\frac{n^k}{b^k} + n^k$$

$$T(n) = a^3T\left(\frac{n}{b^3}\right) + a^2\frac{n^k}{b^{2k}} + a\frac{n^k}{b^k} + n^k$$

2. General Form After i Steps

From the steps above, we can generalize the recurrence for i iterations:

$$T(n) = a^iT\left(\frac{n}{b^i}\right) + n^k \sum_{j=0}^{i-1} \left(\frac{a}{b^k}\right)^j$$

3. Reaching the Base Case

The substitution stops when the subproblem size $\frac{n}{b^i} = 1$, which implies:

$$i = \log_b n$$

Assuming the base case $T(1) = c$, we substitute i back into the general form:

$$T(n) = a^{\log_b n} \cdot c + n^k \sum_{j=0}^{\log_b n - 1} \left(\frac{a}{b^k}\right)^j$$

Using the identity $a^{\log_b n} = n^{\log_b a}$:

$$T(n) = c \cdot n^{\log_b a} + n^k \sum_{j=0}^{\log_b n - 1} \left(\frac{a}{b^k}\right)^j$$